

Q-SENTINEL D1: Mathematical Model and Protocol Specification

Quantum / Detection Workstream

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Abstract

This document is the mathematical specification for the D1–D6 Q-SENTINEL prototype. It fixes the six-state Pauli alphabet, Bell-pair resource, teleportation measurement/correction map, verification predicate, basis-guessing forgery bound, Hoeffding/Chernoff calibration quantities, and the observables used by the six deterministic detectors. The probability statements are proved under the explicit independent-round and basis-guessing model. A real security proof for a complete deployed QDS protocol would additionally require an authenticated classical channel, a composable security definition, and independent peer review.

1 Notation and State Alphabet

Let $\mathcal{B} = \{X, Y, Z\}$ denote the Pauli measurement bases. The six allowed public-key states are

$$\begin{aligned} Z : & |0\rangle, |1\rangle, \\ X : & |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}, \\ Y : & |+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}. \end{aligned}$$

For each round $j \in \{1, \dots, n\}$, a basis $B_j \in \mathcal{B}$ and an associated eigenstate label E_j are generated from a classical seed. The expected projective outcome is denoted $e_j \in \{0, 1\}$, with plus/0 eigenstates mapped to 0 and minus/1 eigenstates mapped to 1.

2 Bell Resource and Teleportation

The shared resource is

$$|\Phi^+\rangle_{BC} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Let the input qubit be

$$|\psi\rangle_A = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The standard Bell-state measurement (BSM) on A, B yields two classical bits $m_1 m_2$. The receiver's state before correction is, up to a global phase,

$$|\psi_C^{\text{raw}}\rangle = P_{m_1 m_2} |\psi\rangle, \tag{1}$$

where

$m_1 m_2$	$P_{m_1 m_2}$
00	I
01	X
10	Z
11	XZ

and the receiver applies the same Pauli operator to undo the BSM-dependent rotation. Therefore

$$P_{m_1 m_2}^\dagger P_{m_1 m_2} |\psi\rangle = |\psi\rangle. \quad (2)$$

Since Pauli operators are unitary, the correction preserves normalization and fidelity.

Lemma 1 (Uniform BSM outcomes). *For a maximally entangled $|\Phi^+\rangle$ resource, each Bell-basis outcome has probability $1/4$, independently of the input pure state.*

Proof. The teleportation decomposition is

$$|\psi\rangle |\Phi^+\rangle = \frac{1}{2} \sum_{k=1}^4 |\text{Bell}_k\rangle P_k |\psi\rangle.$$

Each branch has amplitude norm $1/2$, hence probability $1/4$. □

3 QDS Key Material and Public-Key Distribution

A seed expands deterministically to

$$K = ((B_1, E_1), \dots, (B_n, E_n)), \quad B_j \in \{X, Y, Z\}.$$

The corresponding eigenstates are teleported through the Bell-pair resource. The software records pair-provenance identifiers and a commitment

$$C = H(\text{verifier_id} \parallel B_1 \parallel \dots \parallel B_n).$$

A signature envelope contains the message digest $H(M)$, commitment C , a fresh nonce, verifier identity, and pair-provenance identifiers.

4 Verification Predicate

For round j , the verifier performs a projective measurement in the committed basis B_j and obtains $M_j \in \{0, 1\}$. Define

$$D = \sum_{j=1}^n \mathbf{1}[M_j \neq e_j], \quad r = \frac{D}{n}.$$

In the noiseless honest model each distributed public-key state is exactly an eigenstate of its verification basis, so

$$D = 0 \quad \text{with probability 1.}$$

The D3 reference implementation therefore accepts an untampered signature if and only if all message, identity, commitment, provenance, freshness, and projective-measurement predicates pass the configured mismatch threshold.

Theorem 1 (Deterministic honest acceptance). *In the ideal noiseless model, with matching message digest, verifier identity, basis commitment, pair provenance and fresh nonce, the honest signature is accepted with probability 1 when the mismatch threshold is zero.*

Proof. Each measurement is performed in the eigenbasis of the distributed state, so $M_j = e_j$ deterministically for every j . Hence $D = 0$. All classical binding predicates are equalities by construction, and freshness holds for a new nonce. Therefore every term in the acceptance predicate is true. □

5 Basis-Guessing Forgery Bound

Assume an attacker does not know the committed basis sequence and that rounds are independent. For a two-basis alphabet $\{X, Z\}$, the attacker chooses the correct basis with probability $1/2$ and otherwise measures in the wrong basis, which produces a random eigenvalue. Even in the wrong-basis case the attacker can guess the expected sign with probability $1/2$, so

$$p_2 = \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{3}{4}.$$

For the six-state alphabet $\{X, Y, Z\}$,

$$p_6 = \frac{1}{3}(1) + \frac{2}{3}\left(\frac{1}{2}\right) = \frac{2}{3}.$$

Theorem 2 (Independent-round forgery bound). *Under the basis-guessing model above, the probability that a forger passes all n rounds satisfies*

$$P_{\text{forge}}(n) \leq p^n, \quad p \in \left\{\frac{3}{4}, \frac{2}{3}\right\}.$$

Proof. The probability of passing a single round is at most p . Independence of the rounds gives the product bound p^n . $\square$

Corollary 1 (Numerical values). *At $n = 128$,*

$$\left(\frac{3}{4}\right)^{128} \approx 1.0182 \times 10^{-16},$$

and

$$\left(\frac{2}{3}\right)^{128} \approx 2.8862 \times 10^{-23}.$$

At $n = 256$ the corresponding values are approximately 1.0368×10^{-32} and 8.3299×10^{-46} . The first 128-round value is the explicit D3 benchmark quoted in the project dossier.

6 Hoeffding and Chernoff Statistical Bounds

Let $X_1, \dots, X_n \in \{0, 1\}$ be mismatch indicators with mean p . Hoeffding's one-sided inequality gives

$$\Pr(\hat{p} - p \geq \varepsilon) \leq e^{-2n\varepsilon^2}.$$

For a target one-sided error budget α ,

$$\varepsilon_\alpha = \sqrt{\frac{\ln(1/\alpha)}{2n}}, \quad p_{\text{th}} = p + \varepsilon_\alpha.$$

The Bernoulli Chernoff–KL form is

$$\Pr(\hat{p} \geq q) \leq \exp[-nD_{\text{KL}}(q||p)],$$

with

$$D_{\text{KL}}(q||p) = q \ln \frac{q}{p} + (1 - q) \ln \frac{1 - q}{1 - p}.$$

The D5 implementation exposes both bounds. For finite-sample validation it also computes the exact binomial operating probability; the exact tail should not be confused with the generally looser concentration upper bound.

7 Channel and Entanglement Observables

Define

$$\text{QBER} = \frac{D}{n}.$$

The Bell-state integrity monitor uses the CHSH value S and Bell-state fidelity

$$F_{\Phi^+} = \langle \Phi^+ | \rho | \Phi^+ \rangle.$$

The classical CHSH limit is $S \leq 2$ and the quantum Tsirelson bound is $S \leq 2\sqrt{2}$. The D4 channel detector combines QBER, a four-outcome χ^2 goodness-of-fit test on BSM counts, a one-sided CUSUM, and Wald’s SPRT.

8 Freshness, Identity and Provenance

Every signature carries a nonce. A consumed nonce cannot be reused. The D4 replay guard maintains an explicit registry. The identity-binding detector checks verifier identity, basis commitment, message binding and Bell-pair provenance. These predicates can be mirrored by the team’s permissioned ledger without importing a learned model into the trust path.

9 Six-Detector Mapping

Detector	Attack / purpose	Primary evidence
D1	Baseline integrity	eigenstate measurement mismatch
D2	Signature forgery	Hoeffding/Chernoff mismatch bound
D3	Entanglement hijack	CHSH and Bell fidelity
D4	Intercept-resend / collective attacks	QBER, χ^2 , CUSUM, SPRT
D5	Replay attacks	nonce freshness
D6	Impersonation / unauthorized verification	identity, commitment, provenance

10 Implementation and Security Boundary

The D2–D6 code is a simulation-first engineering artifact. Its deterministic rules can be audited and reproduced, but the repository does not claim a composable end-to-end security proof for all protocol environments. A publication-grade QDS security proof must additionally specify the authenticated classical channel, adversary capabilities, key distribution assumptions, finite sample effects, noise model, multi-verifier composition, and a formal security notion. Those assumptions are deliberately visible here rather than hidden.